

- 1 The first atomic bomb explosion was during the Trinity Test in 1945. Based on a series of photographs of the explosion, along with a size scale and time stamps, British physicist Sir Geoffrey Ingram Taylor was able to estimate the power released by the explosion.

The radius  $R$  of the fireball at time  $t$  after the explosion was found to be related to the energy  $E$  that was released and the density  $\rho$  of the surrounding medium as follows:

$$R = E^{0.2} \rho^{-0.2} t^Z$$

where  $Z$  is a constant.

- (a) Show that the value of  $Z$  is 0.4.

$$[R] = [E^{0.2} \rho^{-0.2} t^Z] \quad [1]$$

$$m = kg^{0.2} m^{0.4} s^{-0.4} kg^{-0.2} m^{0.6} s^Z \quad [1]$$

$$Z = 0.4$$

$$Z = \dots\dots\dots [2]$$

- (b) Given that the radius  $R$  is  $(80 \pm 2)$  m at time  $t$  being  $(0.006 \pm 0.001)$  s with density of air being  $(1.2 \pm 0.1)$  kg m<sup>-3</sup>, determine the energy  $E$  released by the bomb with its associated uncertainty.

$$R = E^{0.2} \rho^{-0.2} t^{0.4}$$

$$E = R^5 \rho t^{-2}$$

$$\frac{\Delta E}{E} = 5 \frac{\Delta R}{R} + \frac{\Delta \rho}{\rho} + 2 \frac{\Delta t}{t} \quad [1]$$

$$E = (80)^5 (1.2) (0.006)^{-2}$$

$$\Delta E = (1.092 \times 10^{14}) \left[ 5 \left( \frac{2}{80} \right) + \frac{0.1}{1.2} + 2 \left( \frac{0.001}{0.006} \right) \right]$$

$$E = 1.09 \times 10^{14} \quad [1]$$

$$\Delta E = 5.92 \times 10^{13} \quad [1]$$

$$E \pm \Delta E = (1.1 \pm 0.6) \times 10^{14} \quad [1]$$

$$E \pm \Delta E = \dots\dots\dots \text{J} [3]$$

- (c) Of all the data collected, state and explain which measurement you would improve to increase the precision of your answer in (b).

$$5 \left( \frac{2}{80} \right) = 0.125 \quad \frac{0.1}{1.2} = 0.08 \quad 2 \left( \frac{0.001}{0.006} \right) = 0.33 \quad [1]$$

Improve  $t$  to increase precision of  $E$ .  
 ..... [2]

Due to its large percentage uncertainty and factor 2 in the calculation of uncertainty of  $E$ , it contributes to the majority of uncertainty of  $E$ . [1]  
 ..... [2]

[Total: 7]